

## FCAI ANU Materials – Semi Seniors – Chapter 3 – Computer Networks

### Chapter 3: Transport Layer - Multiple Choice Questions

#### Questions 1-10

**1. What is the primary role of the transport layer?**

- a) To provide communication services directly to application processes running on different hosts.
- b) To route packets between different networks.
- c) To define the electrical specifications for data transmission.
- d) To compress application data for efficient transfer.

**Answer: a**

*Explanation: As stated on Page 3, the transport layer's critical role is "providing communication services directly to the application processes running on different hosts."*

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**2. The two main transport-layer protocols in the Internet are:**

- a) IP and HTTP
- b) TCP and UDP
- c) FTP and SMTP
- d) Ethernet and PPP

**Answer: b**

*Explanation: Page 7 explicitly states: "the Internet has two protocols—TCP and UDP."*

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**3. Logical communication between processes running on different hosts is provided by the:**

- a) Network Layer
- b) Data Link Layer
- c) Transport Layer
- d) Physical Layer

**Answer: c**

*Explanation: Page 5 defines that a "transport-layer protocol provides for logical communication between application processes running on different hosts."*

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**4. On the sending side, the transport layer converts application messages into:**

- a) Datagrams
- b) Frames
- c) Segments
- d) Bits

**Answer: c**

*Explanation: Page 6 explains that the transport layer converts messages into "transport-layer packets, known as transport-layer segments in Internet terminology."*

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**5. Which of the following statements is TRUE regarding network routers and transport-layer segments?**

- a) Routers examine the transport-layer header fields to make routing decisions.
- b) Routers act only on the network-layer fields of the datagram.
- c) Routers use port numbers to forward packets efficiently.
- d) Routers reassemble transport-layer segments before forwarding them.

**Answer: b**

*Explanation: Page 6 notes: "network routers act only on the network-layer fields of the datagram; that is, they do not examine the fields of the transport-layer segment."*

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**6. The services provided by a transport-layer protocol can be constrained by the:**

- a) Services of the application-layer protocol.
- b) Services of the underlying network-layer protocol.
- c) Speed of the CPU in the end systems.
- d) Type of physical media used.

**Answer: b**

*Explanation: Page 9 states: "the services that a transport protocol can provide are often constrained by the service model of the underlying network-layer protocol."*

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**7. A transport protocol can offer reliable data transfer even when the underlying network protocol is unreliable because:**

- a) The network layer guarantees delivery of all packets.
- b) The transport layer can implement mechanisms like acknowledgments and retransmissions.

- c) Application processes handle their own error correction.
- d) Routers correct errors in transport-layer segments.

**Answer: b**

*Explanation: Page 10 gives this as an example: "a transport protocol can offer reliable data transfer service to an application even when the underlying network protocol is unreliable."*

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**8. The Internet's network-layer protocol, IP, provides what kind of service model?**

- a) Guaranteed delivery
- b) Best-effort delivery
- c) Constant bit-rate delivery
- d) Secure and encrypted delivery

**Answer: b**

*Explanation: Page 12 describes the IP service model as a "best-effort delivery service."*

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**9. Which of the following is NOT a characteristic of IP's service model?**

- a) It does not guarantee segment delivery.
- b) It does not guarantee orderly delivery of segments.
- c) It does not guarantee the integrity of the data.
- d) It provides congestion control to prevent network overload.

**Answer: d**

*Explanation: Page 12 lists what IP does not guarantee (delivery, order, integrity). Congestion control is a service provided by TCP at the transport layer, not IP.*

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**10. The fundamental responsibility of both UDP and TCP is to extend delivery from:**

- a) Process-to-process to host-to-host.
- b) Host-to-host to process-to-process.
- c) Bit-to-bit to segment-to-segment.
- d) Socket-to-socket to application-to-application.

**Answer: b**

*Explanation: Page 13 states: "The most fundamental responsibility of UDP and TCP is to extend IP's delivery service between two end systems to a delivery service between two processes running on the end systems."*

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## Questions 11-20

**11. The process of delivering data from the network layer to the correct application process is known as:**

- a) Multiplexing
- b) Demultiplexing
- c) Encapsulation
- d) Decapsulation

**Answer: b**

*Explanation: Page 19 defines demultiplexing as "the job of delivering the data in a transport-layer segment to the correct socket."*

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**12. The job of gathering data from different sockets and adding headers to create segments is called:**

- a) Demultiplexing
- b) Multiplexing
- c) Encapsulation
- d) Fragmentation

**Answer: b**

*Explanation: Page 19 defines multiplexing as gathering data from sockets, encapsulating it with headers, and passing segments to the network layer.*

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**13. In a transport-layer segment, the fields used to identify the destination application are the:**

- a) Sequence number and acknowledgment number fields.
- b) Source and destination IP address fields.
- c) Source and destination port number fields.
- d) Checksum and length fields.

**Answer: c**

*Explanation: Page 20 indicates that the "source port number field" and the "destination port number field" are the special fields used for multiplexing/demultiplexing.*

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**14. Port numbers in the range of 0 to 1023 are known as:**

- a) Ephemeral port numbers
- b) Well-known port numbers
- c) Registered port numbers
- d) Dynamic port numbers

**Answer: b**

*Explanation: Page 20 states: "The port numbers ranging from 0 to 1023 are called well-known port numbers."*

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**15. How does a host implement demultiplexing for a UDP segment?**

- a) It uses the combination of source and destination IP addresses and port numbers.
- b) It uses only the destination IP address.
- c) It uses only the destination port number.
- d) It uses the sequence number in the segment.

**Answer: c**

*Explanation: Page 22-24 explains that for UDP, the destination host directs the segment to the socket by examining the destination port number alone.*

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**16. A TCP socket is identified by a:**

- a) Destination port number
- b) Source port number and destination port number
- c) Four-tuple: (source IP, source port, destination IP, destination port)
- d) Sequence number

**Answer: c**

*Explanation: Page 26 clearly states: "a TCP socket is identified by a four-tuple: (source IP address, source port number, destination IP address, destination port number)."*

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**17. A Web server can distinguish between multiple simultaneous connections from the same client host because:**

- a) The client uses different destination port numbers for each connection.
- b) The server assigns a different IP address for each connection.
- c) The client can use different source port numbers for each connection.
- d) The TCP segments have different sequence numbers.

**Answer: c**

*Explanation: The example on Page 27 shows that a client can assign different source port numbers for its multiple connections to the same server port (e.g., port 80), allowing the server to demultiplex them.*

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**18. Which transport protocol is connectionless?**

- a) TCP
- b) UDP
- c) Both TCP and UDP
- d) Neither TCP nor UDP

**Answer: b**

*Explanation: Page 34 states: "with UDP there is no handshaking... For this reason, UDP is said to be connectionless."*

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**19. Why might an application developer choose UDP over TCP?**

- a) For reliable data transfer.
- b) For finer application-level control over data transmission timing.
- c) For built-in congestion control.
- d) To ensure in-order delivery of all packets.

**Answer: b**

*Explanation: Page 35 lists "Finer application-level control over what data is sent, and when" as a primary reason for choosing UDP.*

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**20. A key reason DNS typically uses UDP is because:**

- a) DNS requires high throughput for large file transfers.
- b) DNS queries and responses are small and UDP avoids connection establishment delay.
- c) DNS needs reliable delivery of all query packets.
- d) DNS uses encryption provided by UDP.

**Answer: b**

*Explanation: Page 36 mentions that "UDP does not introduce any delay to establish a connection. This is probably the principal reason why DNS runs over UDP."*

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## Questions 21-30

**21. Which of the following is an advantage of UDP over TCP?**

- a) Connection state maintenance
- b) Congestion control
- c) Smaller packet header overhead
- d) Reliable data transfer

**Answer: c**

*Explanation: Page 37 notes that "TCP segment has 20 bytes of header overhead... whereas UDP has only 8 bytes."*

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**22. According to Figure 3.6, which application typically uses UDP?**

- a) Electronic Mail (SMTP)
- b) File Transfer (FTP)
- c) Remote Terminal Access (Telnet)
- d) Name Translation (DNS)

**Answer: d**

*Explanation: Figure 3.6 on Page 38 lists "Name translation" with the underlying transport protocol "Typically UDP."*

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**23. A significant problem with running multimedia applications over UDP is:**

- a) The 20-byte header is too large.
- b) It provides reliable data transfer, which is unnecessary.
- c) It has no congestion control, which can lead to network collapse.
- d) It requires a three-way handshake, causing delay.

**Answer: c**

*Explanation: Page 40 discusses this problem in detail, stating that "the lack of congestion control in UDP can result in high loss rates... and the crowding out of TCP sessions."*

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**24. It is possible to have reliable data transfer when using UDP by:**

- a) Relying on the network layer to retransmit lost packets.
- b) Building reliability (acknowledgments and retransmissions) into the application itself.
- c) Using the checksum field to correct bit errors.

- d) It is impossible; reliable transfer requires TCP.

**Answer: b**

*Explanation: Page 41 states: "it is possible for an application to have reliable data transfer when using UDP. This can be done if reliability is built into the application itself."*

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**25. The UDP header consists of how many fields, each of 2 bytes?**

- a) Two
- b) Four
- c) Six
- d) Eight

**Answer: b**

*Explanation: Page 43 states: "The UDP header has only four fields, each consisting of two bytes."*

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**26. The UDP checksum is used for:**

- a) Error correction
- b) Error detection
- c) Data compression
- d) Flow control

**Answer: b**

*Explanation: Page 44 states: "The UDP checksum provides for error detection."*

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**27. Why does UDP provide error detection at the transport layer when link-layer protocols may also provide it?**

- a) To reduce the processing load on routers.
- b) Because there is no guarantee that all links provide error checking, and errors can be introduced in a router's memory.
- c) Because link-layer checksums are always incorrect.
- d) To provide faster error correction.

**Answer: b**

*Explanation: Page 44 explains this rationale: "there is no guarantee that all the links between source and destination provide error checking" and "bit errors could be introduced when a segment is stored in a router's memory."*

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**28. The service abstraction provided by a reliable data transfer protocol is that of a:**

- a) Best-effort channel
- b) Reliable channel
- c) Connectionless channel
- d) Multiplexed channel

**Answer: b**

*Explanation: Page 48 states: "The service abstraction provided to the upper-layer entities is that of a reliable channel."*

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**29. In the rdt protocol, the function `udt_send()` is used to:**

- a) Deliver data to the upper layer.
- b) Send packets where the underlying channel is unreliable.
- c) Acknowledge correctly received packets.
- d) Check for errors in packets.

**Answer: b**

*Explanation: Page 50 explains that both sides send packets by calling `udt_send()` where "udt stands for unreliable data transfer."*

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**30. The simplest protocol, `rdt1.0`, is designed for a channel that is:**

- a) Lossy and has bit errors.
- b) Completely reliable.
- c) Can reorder packets.
- d) Has a variable delay.

**Answer: b**

*Explanation: Page 51 introduces `rdt1.0` as a protocol "for a completely reliable channel."*

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## Questions 31-40

**31. In the message-dictation analogy, the receiver saying "OK" is similar to a:**

- a) Negative Acknowledgment (NAK)
- b) Positive Acknowledgment (ACK)

- c) Sequence Number
- d) Timer

**Answer: b**

*Explanation: Page 55 describes this scenario: "the message taker might say 'OK' after each sentence... These control messages... positive acknowledgments ('OK')".*

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**32. Protocols that use retransmission to provide reliable data transfer are known as:**

- a) FDM protocols
- b) TDM protocols
- c) ARQ protocols
- d) CSMA protocols

**Answer: c**

*Explanation: Page 56 states: "reliable data transfer protocols based on such retransmission are known as ARQ (Automatic Repeat reQuest) protocols."*

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**33. What are the three additional capabilities required for ARQ protocols?**

- a) Encryption, Compression, Multiplexing
- b) Error detection, Receiver feedback, Retransmission
- c) Routing, Forwarding, Addressing
- d) Synchronization, Sequencing, Flow Control

**Answer: b**

*Explanation: Page 56 lists them: "Error detection," "Receiver feedback," and "Retransmission."*

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**34. In rdt2.0, what fatal flaw is identified?**

- a) It cannot handle bit errors.
- b) It assumes ACK and NAK packets are never corrupted.
- c) It has no sequence numbers.
- d) It is too slow.

**Answer: b**

*Explanation: Page 61 points out the flaw: "we haven't accounted for the possibility that the ACK or NAK packet could be corrupted!"*

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**35. How is the problem of corrupted ACK/NAK packets solved in rdt2.1 and rdt2.2?**

- a) By using a stronger checksum.
- b) By adding a sequence number to data packets.
- c) By increasing the timer interval.
- d) By using a faster transmission medium.

**Answer: b**

*Explanation: Page 63 explains the solution: "add a new field to the data packet and have the sender number its data packets by putting a sequence number into this field."*

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**36. For a stop-and-wait protocol, a sequence number of how many bits is sufficient?**

- a) 0 bits
- b) 1 bit
- c) 16 bits
- d) 32 bits

**Answer: b**

*Explanation: Page 63 states: "For this simple case of a stop-and-wait protocol, a 1-bit sequence number will suffice."*

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**37. The key difference between rdt2.1 and rdt2.2 is that rdt2.2:**

- a) Uses a 2-bit sequence number.
- b) Does not use NAKs; instead, it sends an ACK for the last correctly received packet.
- c) Uses a timer for the first time.
- d) Is designed for a lossy channel.

**Answer: b**

*Explanation: Page 66 explains that in rdt2.2, "We can accomplish the same effect as a NAK if, instead of sending a NAK, we send an ACK for the last correctly received packet."*

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**38. In rdt3.0, the problem of lost packets is addressed by:**

- a) Using longer sequence numbers.
- b) Implementing a countdown timer and retransmitting on timeout.
- c) Using more powerful error-correcting codes.
- d) Increasing the transmission power.

**Answer: b**

*Explanation: Page 70-72 discuss how rdt3.0 handles packet loss using a "countdown timer" and retransmission when the timer expires.*

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**39. rdt3.0 is often called the:**

- a) Sliding-window protocol.
- b) Go-Back-N protocol.
- c) Selective Repeat protocol.
- d) Alternating-bit protocol.

**Answer: d**

*Explanation: Page 76 states: "protocol rdt3.0 is sometimes known as the alternating-bit protocol."*

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**40. The main performance problem with rdt3.0 (stop-and-wait) is:**

- a) High header overhead.
- b) Low sender utilization.
- c) Inability to handle bit errors.
- d) Complex receiver implementation.

**Answer: b**

*Explanation: Page 77-81 discuss the "performance problem" of stop-and-wait, leading to a "rather dismal sender utilization."*

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## **Questions 41-50**

**41. The solution to the performance problem of stop-and-wait protocols is:**

- a) Using a larger sequence number space.
- b) Pipelining
- c) Increasing the transmission rate R.
- d) Decreasing the round-trip time RTT.

**Answer: b**

*Explanation: Page 82 states: "The solution to this particular performance problem is simple... the sender is allowed to send multiple packets without waiting for acknowledgments... this technique is known as pipelining."*

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**42. Pipelining requires which of the following? (Select all that apply)**

- a) An increased range of sequence numbers.
- b) The sender and receiver to buffer more than one packet.
- c) A change in the underlying physical layer.
- d) Different error-recovery approaches like Go-Back-N or Selective Repeat.

**Answer: a, b, d**

*Explanation: Page 84 lists the consequences of pipelining, which include all of these except (c).*

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**43. In a Go-Back-N (GBN) protocol, the sender is allowed to have up to N:**

- a) Acknowledged packets.
- b) Packets received by the application layer.
- c) Unacknowledged packets in the pipeline.
- d) Total packets in the entire transmission.

**Answer: c**

*Explanation: Page 85 defines GBN: "the sender is allowed to transmit multiple packets... but is constrained to have no more than some maximum allowable number, N, of unacknowledged packets in the pipeline."*

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**44. In GBN, the window size N is also referred to as the:**

- a) Congestion window.
- b) Receive window.
- c) Sliding window.
- d) Buffer window.

**Answer: c**

*Explanation: Page 88 states: "N is often referred to as the window size and the GBN protocol itself as a sliding-window protocol."*

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**45. In GBN, when a timeout occurs, the sender:**

- a) Retransmits only the packet that caused the timeout.
- b) Retransmits all packets that have been sent but not yet acknowledged.
- c) Retransmits the last packet that was acknowledged.

- d) Stops transmitting and sends a reset signal.

**Answer: b**

*Explanation: Page 93 describes the sender's behavior on a timeout: "the sender resends all packets that have been previously sent but that have not yet been acknowledged."*

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**46. How does a GBN receiver respond to an out-of-order packet that is error-free?**

- a) It buffers the packet for later delivery.
- b) It sends a negative acknowledgment (NAK).
- c) It discards the packet and resends an ACK for the last in-order packet.
- d) It delivers the packet immediately to the application layer.

**Answer: c**

*Explanation: Page 94 states: "In all other cases [i.e., not the expected in-order packet], the receiver discards the packet and resends an ACK for the most recently received in-order packet."*

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**47. The main advantage of the GBN receiver's approach of discarding out-of-order packets is:**

- a) Higher throughput.
- b) Simplicity (the receiver need not buffer out-of-order packets).
- c) Lower delay.
- d) Better error correction.

**Answer: b**

*Explanation: Page 95 explains the advantage: "the simplicity of receiver buffering—the receiver need not buffer any out-of-order packets."*

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**48. The Selective Repeat (SR) protocol aims to solve which problem of GBN?**

- a) The complexity of the receiver.
- b) The need for sequence numbers.
- c) Unnecessary retransmissions when the window size and bandwidth-delay product are large.
- d) The lack of flow control.

**Answer: c**

*Explanation: Page 98 explains the motivation for SR: "A single packet error can thus cause GBN*

*to retransmit a large number of packets, many unnecessarily... selective-repeat protocols avoid unnecessary retransmissions."*

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**49. In SR, the receiver acknowledges:**

- a) Only the last in-order packet received.
- b) All packets that have been received correctly, whether in or out of order.
- c) Only packets that are in order.
- d) Only packets that are out of order.

**Answer: b**

*Explanation: Page 98 states that SR requires the receiver to "individually acknowledge correctly received packets." This implies acknowledging correct packets even if they are out-of-order.*

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**50. TCP is considered a connection-oriented protocol because:**

- a) It uses IP datagrams for transmission.
- b) It provides a reliable data transfer service.
- c) The two processes must first "handshake" before data transfer.
- d) It uses sequence numbers for its segments.

**Answer: c**

*Explanation: Page 102 states: "TCP is said to be connection-oriented because before one application process can begin to send data to another, the two processes must first 'handshake' with each other."*

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## **Questions 51-60**

**51. TCP connection state is maintained in:**

- a) The end systems only.
- b) The intermediate routers only.
- c) Both end systems and intermediate routers.
- d) The network layer.

**Answer: a**

*Explanation: Page 102 makes it clear: "the TCP protocol runs only in the end systems and not in the intermediate network elements... the intermediate routers are completely oblivious to TCP connections."*

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**52. TCP provides what kind of communication service?**

- a) Half-duplex
- b) Simplex
- c) Full-duplex
- d) Multicast

**Answer: c**

*Explanation: Page 103 states: "A TCP connection provides a full-duplex service."*

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**53. The procedure for establishing a TCP connection is known as:**

- a) A two-way handshake.
- b) A three-way handshake.
- c) A four-way handshake.
- d) Connection forwarding.

**Answer: b**

*Explanation: Page 105 states: "this connection-establishment procedure is often referred to as a three-way handshake."*

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**54. During the TCP three-way handshake, the first segment has which flag bit set?**

- a) ACK
- b) SYN
- c) FIN
- d) RST

**Answer: b**

*Explanation: Page 151 describes step 1: "the SYN bit, is set to 1. For this reason, this special segment is referred to as a SYN segment."*

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**55. The TCP send buffer is used to:**

- a) Store data that has been received from the network and is ready for the application to read.
- b) Store data that the application has written and is ready to be sent.
- c) Store acknowledgments received from the other end.

- d) Store out-of-order segments until the missing data arrives.

**Answer: b**

*Explanation: Figure 3.28 on Page 107 shows data flowing from the application, through the socket, into the "TCP send buffer," from where TCP sends segments.*

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**56. The Maximum Segment Size (MSS) is the:**

- a) Maximum size of a TCP segment including headers.
- b) Maximum amount of application-layer data in a segment.
- c) Maximum size of the IP datagram.
- d) Maximum size of the send and receive buffers.

**Answer: b**

*Explanation: Page 108 clarifies: "the MSS is the maximum amount of application-layer data in the segment, not the maximum size of the TCP segment including headers."*

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**57. A typical MSS value over an Ethernet network is:**

- a) 1500 bytes
- b) 1460 bytes
- c) 40 bytes
- d) 65535 bytes

**Answer: b**

*Explanation: Page 108 states: "a typical value of MSS is 1460 bytes."*

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**58. Which field in the TCP header is NOT also in the UDP header?**

- a) Source port number
- b) Destination port number
- c) Checksum
- d) Acknowledgment number

**Answer: d**

*Explanation: Comparing the UDP segment structure (Page 42) and the TCP segment structure (Page 111), the Acknowledgment number is a field unique to TCP.*

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**59. The TCP header length field is necessary because the TCP header:**

- a) Is always 20 bytes long.
- b) Can be of variable length due to the options field.
- c) Includes a data field of variable length.
- d) Is shorter than the UDP header.

**Answer: b**

*Explanation: Page 112 states: "The TCP header can be of variable length due to the TCP options field... The 4-bit header length field specifies the length of the TCP header."*

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**60. The TCP flag bit that is used to acknowledge received segments is the:**

- a) SYN bit
- b) ACK bit
- c) FIN bit
- d) RST bit

**Answer: b**

*Explanation: Page 113 states: "The ACK bit is used to indicate that the value carried in the acknowledgment field is valid."*

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## Questions 61-70

**61. In a TCP segment, the sequence number is the byte-stream number of the:**

- a) Last byte in the segment.
- b) First byte in the segment.
- c) The byte being acknowledged.
- d) The next byte expected by the receiver.

**Answer: b**

*Explanation: Page 115 states: "The sequence number for a segment is therefore the byte-stream number of the first byte in the segment."*

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**62. TCP acknowledgments are cumulative, meaning that an acknowledgment number 'n' indicates that:**

- a) Segment 'n' has been received correctly.
- b) All bytes up to and including byte 'n-1' have been received correctly.

- c) All bytes up to and including byte 'n' have been received correctly.
- d) Byte 'n' is the next byte expected.

**Answer: b**

*Explanation: The text on Page 118 explains that if a host is waiting for byte 536, it means it has received all bytes up through 535. Therefore, an ACK for 536 acknowledges all bytes up to 535.*

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**63. If a TCP receiver receives segments out of order, a typical implementation will:**

- a) Always discard them.
- b) Always buffer them until the missing data arrives.
- c) Send a NAK for the missing segment.
- d) Reset the connection.

**Answer: b**

*Explanation: Page 119 states that while the RFCs leave it up to the programmer, "the latter choice [keeping the out-of-order bytes and waiting for the missing bytes] is more efficient in terms of network bandwidth, and is the approach taken in practice."*

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**64. TCP estimates the Round-Trip Time (RTT) by:**

- a) Taking a single SampleRTT for all transmitted segments.
- b) Taking a SampleRTT for every segment transmitted.
- c) Taking one SampleRTT measurement at a time for a transmitted, unacknowledged segment.
- d) Using a fixed value defined in the RFCs.

**Answer: c**

*Explanation: Page 121 states: "most TCP implementations take only one SampleRTT measurement at a time. That is, at any point in time, the SampleRTT is being estimated for only one of the transmitted but currently unacknowledged segments."*

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**65. The formula for EstimatedRTT is:  $\text{EstimatedRTT} = (1 - \alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}$ . The recommended value for  $\alpha$  is:**

- a) 0.1
- b) 0.125
- c) 0.25
- d) 0.5

**Answer: b**

*Explanation: Page 122 gives the formula and states: "The recommended value of  $\alpha$  is  $\alpha = 0.125$  (that is,  $1/8$ )."*

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**66. The DevRTT is an estimate of the:**

- a) Average RTT.
- b) Variability of the RTT.
- c) Minimum RTT.
- d) Maximum RTT.

**Answer: b**

*Explanation: Page 124 defines DevRTT: "an estimate of how much SampleRTT typically deviates from EstimatedRTT."*

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**67. How is TCP's retransmission timeout interval (TimeoutInterval) calculated?**

- a)  $\text{TimeoutInterval} = \text{EstimatedRTT}$
- b)  $\text{TimeoutInterval} = \text{EstimatedRTT} + \text{DevRTT}$
- c)  $\text{TimeoutInterval} = \text{EstimatedRTT} + 4 * \text{DevRTT}$
- d)  $\text{TimeoutInterval} = 2 * \text{EstimatedRTT}$

**Answer: c**

*Explanation: Page 125 provides the formula: " $\text{TimeoutInterval} = \text{EstimatedRTT} + 4 \cdot \text{DevRTT}$ "*

---

**68. TCP's reliable data transfer service ensures that the data stream is: (Select all that apply)**

- a) Uncorrupted.
- b) Without gaps.
- c) Without duplication.
- d) In sequence.

**Answer: a, b, c, d**

*Explanation: Page 126 states that TCP's reliable data transfer ensures the byte stream is "uncorrupted, without gaps, without duplication, and in sequence."*

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**69. How many retransmission timers does the recommended TCP timer management procedure use?**

- a) One timer for the entire connection.
- b) One timer per transmitted segment.
- c) Two timers: one for sending and one for receiving.
- d) A dynamically changing number of timers.

**Answer: a**

*Explanation: Page 127 states: "the recommended TCP timer management procedures [RFC 6298] use only a single retransmission timer."*

---

**70. When a TCP timeout occurs, the sender retransmits the segment and:**

- a) Stops the timer.
- b) Restarts the timer.
- c) Doubles the timeout interval.
- d) Halves the timeout interval.

**Answer: b**

*Explanation: Page 129 states: "TCP responds to the timeout event by retransmitting the segment that caused the timeout. TCP then restarts the timer."*

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## Questions 71-80

**71. After a timeout, TCP often doubles the timeout interval. This is a form of:**

- a) Flow control.
- b) Congestion control.
- c) Error detection.
- d) Multiplexing.

**Answer: b**

*Explanation: Page 137 states: "This modification provides a limited form of congestion control."*

---

**72. A "duplicate ACK" is an ACK that:**

- a) Acknowledges a new, previously unacknowledged segment.
- b) Reacknowledges a segment for which an ACK has already been sent.
- c) Is received out of order.

- d) Has an incorrect checksum.

**Answer: b**

*Explanation: Page 138 defines a duplicate ACK: "a duplicate ACK is an ACK that reacknowledges a segment for which the sender has already received an earlier acknowledgment."*

---

**73. TCP's "fast retransmit" mechanism is triggered when the sender receives:**

- a) A single duplicate ACK.
- b) Two duplicate ACKs.
- c) Three duplicate ACKs.
- d) A NAK from the receiver.

**Answer: c**

*Explanation: Page 139 states: "If the TCP sender receives three duplicate ACKs for the same data, it takes this as an indication that the segment... has been lost," and performs a fast retransmit.*

---

**74. TCP's error-recovery mechanism is best described as a hybrid of:**

- a) rdt1.0 and rdt2.0.
- b) UDP and IP.
- c) Go-Back-N and Selective Repeat.
- d) Stop-and-wait and Pipelining.

**Answer: c**

*Explanation: Page 143 concludes: "TCP's error-recovery mechanism is probably best categorized as a hybrid of GBN and SR protocols."*

---

**75. The TCP service that matches the sender's rate to the receiver's application read rate is called:**

- a) Congestion Control
- b) Flow Control
- c) Error Control
- d) Multiplexing

**Answer: b**

*Explanation: Page 144 defines flow control: "Flow control is thus a speed-matching service—*

*matching the rate at which the sender is sending against the rate at which the receiving application is reading."*

---

**76. The variable maintained by the TCP sender for flow control is the:**

- a) Congestion window (cwnd)
- b) Receive window (rwnd)
- c) Sequence number
- d) Advertised window

**Answer: b**

*Explanation: Page 145 states: "TCP provides flow control by having the sender maintain a variable called the receive window." The sender's understanding of the receiver's spare buffer space is called rwnd.*

---

**77. The receive window (rwnd) is calculated as:**

- a)  $rwnd = \text{LastByteSent} - \text{LastByteAcked}$
- b)  $rwnd = \text{RcvBuffer} - (\text{LastByteRcvd} - \text{LastByteRead})$
- c)  $rwnd = \text{LastByteRead} - \text{LastByteRcvd}$
- d)  $rwnd = \text{RcvBuffer}$

**Answer: b**

*Explanation: Page 147 provides the formula: " $rwnd = \text{RcvBuffer} - [\text{LastByteRcvd} - \text{LastByteRead}]$ "*

---

**78. To avoid a deadlock when the receiver's window is zero, the TCP sender will:**

- a) Stop sending any segments.
- b) Send a segment with the RST bit set.
- c) Send segments with one data byte periodically.
- d) Double the congestion window.

**Answer: c**

*Explanation: Page 150 describes the solution: "the TCP specification requires Host A to continue to send segments with one data byte when B's receive window is zero."*

---

**79. Which flag bit is set in the first segment of the TCP connection teardown process?**

- a) SYN
- b) ACK
- c) FIN
- d) RST

**Answer: c**

*Explanation: Page 157 describes connection teardown: "the client TCP to send a special TCP segment... This special segment has a flag bit... the FIN bit... set to 1."*

---

**80. After sending its final ACK in the connection teardown, the client TCP enters the:**

- a) FIN\_WAIT\_1 state
- b) FIN\_WAIT\_2 state
- c) TIME\_WAIT state
- d) CLOSED state

**Answer: c**

*Explanation: Page 160 describes the client's state transition: "the client TCP acknowledges the server's segment and enters the TIME\_WAIT state."*

---

## Questions 81-90

**81. The TIME\_WAIT state exists to:**

- a) Allow the server to free its resources faster.
- b) Let the client resend the final ACK in case it is lost.
- c) Slow down the rate of new connection requests.
- d) Ensure all data has been delivered to the application.

**Answer: b**

*Explanation: Page 160 states: "The TIME\_WAIT state lets the TCP client resend the final acknowledgment in case the ACK is lost."*

---

**82. If a host receives a TCP segment for a port that is not open, it will typically send a segment with which flag set?**

- a) SYN
- b) ACK
- c) FIN
- d) RST

**Answer: d**

*Explanation: Page 162 explains this scenario: "the host will send a special reset segment to the source. This TCP segment has the RST flag bit... set to 1."*

---

**83. The initial sequence number in a TCP connection is:**

- a) Always 0.
- b) Always 1.
- c) Randomly chosen.
- d) Determined by the MSS.

**Answer: c**

*Explanation: Page 151 states that the client "randomly chooses an initial sequence number (client\_isn)."*

---

**84. Which of the following is a reason for imposing a window size limit N in a protocol like GBN?**

- a) To increase the sequence number space.
- b) For flow control and congestion control.
- c) To simplify the receiver's FSM.
- d) To ensure packets are delivered in order.

**Answer: b**

*Explanation: Page 88 mentions that flow control (Section 3.5) and congestion control (Section 3.7) are reasons to impose a limit.*

---

**85. The event that causes the rdt1.0 sender FSM to transition is:**

- a) rdt\_rcv(packet) from below.
- b) A timeout.
- c) rdt\_send(data) from above.
- d) Receiving an ACK.

**Answer: c**

*Explanation: The FSM for the rdt1.0 sender in Figure 3.9(a) on Page 52 has only one event: rdt\_send(data) from above.*

---

**86. In the context of reliable data transfer, a cumulative acknowledgment:**

- a) Acknowledges only a specific segment.
- b) Acknowledges all segments up to a specified sequence number.
- c) Is sent only when a segment is lost.
- d) Is another name for a NAK.

**Answer: b**

*Explanation: The concept is introduced in GBN (Page 94) and is fundamental to TCP. An ACK for sequence number  $n$  cumulatively acknowledges all data up to  $n-1$ .*

---

**87. The term "segment" is used in this text to refer to:**

- a) Both TCP and UDP packets at the transport layer.
- b) Only TCP packets.
- c) Only UDP packets.
- d) Network-layer packets.

**Answer: a**

*Explanation: Page 11 explains the terminology choice: "we believe that it is less confusing to refer to both TCP and UDP packets as segments, and reserve the term datagram for the network-layer packet."*

---

**88. Which transport protocol does not offer congestion control?**

- a) TCP
- b) UDP
- c) Both TCP and UDP
- d) Neither TCP nor UDP

**Answer: b**

*Explanation: Page 14 states that "TCP also provides congestion control," while Page 40 discusses the "lack of congestion control in UDP."*

---

**89. The primary purpose of the transport layer is to provide communication between \_\_\_\_\_, while the network layer provides communication between \_\_\_\_\_.**

- a) hosts, processes
- b) processes, hosts
- c) networks, links

- d) links, networks

**Answer: b**

*Explanation: This fundamental distinction is made on Page 8: "a transport-layer protocol provides logical communication between processes... a network-layer protocol provides logical-communication between hosts."*

---

**90. In the rdt2.2 receiver FSM (Figure 3.14), what action is taken when a corrupt packet is received?**

- a) Send a NAK.
- b) Send an ACK for the last correctly received in-order packet.
- c) Do nothing.
- d) Reset the connection.

**Answer: b**

*Explanation: rdt2.2 is NAK-free. The receiver FSM in Figure 3.14 shows that upon receiving a corrupt packet, the action is to send an ACK for the last correctly received packet (e.g., `sndpkt=make_pkt(ACK, 1, checksum)`).*

---

## Questions 91-100

**91. The alternating-bit protocol is another name for:**

- a) rdt2.2
- b) rdt3.0
- c) GBN
- d) SR

**Answer: b**

*Explanation: Page 76 states: "protocol rdt3.0 is sometimes known as the alternating-bit protocol."*

---

**92. The transmission delay for a packet of length L bits over a link of rate R bps is:**

- a)  $L / R$
- b)  $R / L$
- c)  $L * R$
- d) RTT

**Answer: a**

*Explanation: Page 79 provides the formula and calculation:  $d_{trans} = L/R$*

---

**93. For a stop-and-wait protocol, the sender utilization is given by:**

- a)  $(L/R) / RTT$
- b)  $(L/R) / (RTT + L/R)$
- c)  $RTT / (L/R)$
- d) 1

**Answer: b**

*Explanation: Page 81 provides the formula for sender utilization in a stop-and-wait protocol:  $U_{sender} = (L/R) / (RTT + L/R)$*

---

**94. In the GBN protocol, the variable base is the sequence number of the:**

- a) Next packet to be sent.
- b) Oldest unacknowledged packet.
- c) Last packet acknowledged.
- d) Last packet sent.

**Answer: b**

*Explanation: Page 87 defines base: "base to be the sequence number of the oldest unacknowledged packet".*

---

**95. In the GBN protocol, the variable nextseqnum is the sequence number of the:**

- a) Next packet to be sent.
- b) Oldest unacknowledged packet.
- c) Last packet acknowledged.
- d) Last packet sent.

**Answer: a**

*Explanation: Page 87 defines nextseqnum: "nextseqnum to be the smallest unused sequence number (that is, the sequence number of the next packet to be sent)".*

---

**96. The sequence number interval [base, nextseqnum-1] in GBN corresponds to packets that have been:**

- a) Sent and acknowledged.
- b) Sent but not yet acknowledged.
- c) Not yet sent.
- d) Received out of order.

**Answer: b**

*Explanation: Page 87 explains: "The interval  $[base, nextseqnum-1]$  corresponds to packets that have been sent but not yet acknowledged."*

---

**97. In Selective Repeat, the sender retransmits:**

- a) All unacknowledged packets when a timeout occurs.
- b) Only the packet that is suspected to be lost.
- c) The packet with the highest sequence number.
- d) No packets; the receiver handles all losses.

**Answer: b**

*Explanation: Page 98 states the key idea of SR: "selective-repeat protocols avoid unnecessary retransmissions by having the sender retransmit only those packets that it suspects were received in error."*

---

**98. A TCP connection is identified by a socket pair at each end. A socket is a combination of:**

- a) IP address and port number.
- b) Sequence number and acknowledgment number.
- c) MAC address and IP address.
- d) Process ID and thread ID.

**Answer: a**

*Explanation: While not explicitly defined in one sentence, the concept is built from Page 17 (sockets as doors) and Page 26 (TCP socket identified by a four-tuple, which is a combination of two IP addresses and two port numbers, i.e., two sockets).*

---

**99. The three-way handshake sequence numbers (client\_isn, server\_isn) are used to:**

- a) Initialize the byte-stream counters for data transfer in each direction.
- b) Determine the MSS for the connection.
- c) Set the initial receive window size.
- d) Encrypt the subsequent data transfer.

**Answer: a**

*Explanation: The handshake establishes the initial sequence numbers for the connection, which are then used to number the data bytes in each direction, as described starting on Page 115.*

---

**100. The TCP segment header field that indicates the number of bytes the receiver is willing to accept is the:**

- a) Sequence Number field.
- b) Acknowledgment Number field.
- c) Receive Window field.
- d) Header Length field.

**Answer: c**

*Explanation: Page 112 states: "The 16-bit receive window field... is used to indicate the number of bytes that a receiver is willing to accept."*

---

## Questions 101-110

**101. The PSH flag in a TCP header is used to:**

- a) Request a higher transmission priority for the segment.
- b) Indicate that the receiver should pass the data to the upper layer immediately.
- c) Signal the start of a three-way handshake.
- d) Reset the connection.

**Answer: b**

*Explanation: Page 113 states: "Setting the PSH bit indicates that the receiver should pass the data to the upper layer immediately."*

---

**102. The URG flag and Urgent Data Pointer in a TCP header are used to:**

- a) Mark data that should be transmitted before all other data.
- b) Correct errors in the segment payload.
- c) Manage flow control.
- d) Handle connection teardown.

**Answer: a**

*Explanation: Page 114 states: "the URG bit is used to indicate that there is data in this segment that the sending-side upper-layer entity has marked as 'urgent'."*

---

**103. When a TCP sender receives three duplicate ACKs, it infers that:**

- a) The network is congestion-free.
- b) The segment following the ACKed segment has been lost.
- c) The receiver's buffer is full.
- d) The connection should be reset.

**Answer: b**

*Explanation: Page 139 explains: "If the TCP sender receives three duplicate ACKs for the same data, it takes this as an indication that the segment following the segment that has been ACKed three times has been lost."*

---

**104. The principle that a TCP sender should not send data faster than the network can handle is the goal of:**

- a) Flow Control
- b) Congestion Control
- c) Error Control
- d) Multiplexing

**Answer: b**

*Explanation: While flow control matches the sender to the receiver, congestion control (introduced on Page 145) matches the sender to the network's capacity.*

---

**105. The principle that a TCP sender should not send data faster than the receiver can process is the goal of:**

- a) Flow Control
- b) Congestion Control
- c) Error Control
- d) Multiplexing

**Answer: a**

*Explanation: Page 144 defines flow control as "matching the rate at which the sender is sending against the rate at which the receiving application is reading."*

---

**106. In the TCP state diagram (Figure 3.41), the state a client is in after sending a SYN and before receiving a SYNACK is:**

- a) CLOSED
- b) SYN\_SENT
- c) SYN\_RCVD
- d) ESTABLISHED

**Answer: b**

*Explanation: Page 159 describes the client's state transition: "After having sent the SYN segment, the client TCP enters the SYN\_SENT state."*

---

**107. In the TCP state diagram (Figure 3.42), the state a server is in after receiving a SYN and sending a SYNACK is:**

- a) LISTEN
- b) SYN\_SENT
- c) SYN\_RCVD
- d) ESTABLISHED

**Answer: c**

*Explanation: Figure 3.42 on Page 161 shows the server transitioning from LISTEN to SYN\_RCVD after "Receive SYN send SYN & ACK".*

---

**108. The initial state for both client and server TCP is:**

- a) LISTEN
- b) SYN\_SENT
- c) CLOSED
- d) ESTABLISHED

**Answer: c**

*Explanation: Page 159 states "The client TCP begins in the CLOSED state." Figure 3.42 also shows the server side starting from CLOSED.*

---

**109. The state where a TCP connection is ready for data transfer is:**

- a) LISTEN
- b) SYN\_SENT
- c) SYN\_RCVD
- d) ESTABLISHED

**Answer: d**

*Explanation: Page 159 states that while in the "ESTABLISHED" state, the TCP client can send and receive TCP segments containing payload data."*

---

**110. The main problem with the GBN protocol that SR solves is:**

- a) The inability to handle bit errors.
- b) The low utilization of the sender.
- c) The need for the receiver to buffer out-of-order packets.
- d) Unnecessary retransmissions when a single packet is lost in a large window.

**Answer: d**

*Explanation: As discussed on Page 98, SR avoids the problem where GBN unnecessarily retransmits many correctly received packets after a single loss.*

---

## Questions 111-120

**111. The `udt_send()` function in the rdt protocols is used to send packets over a channel that is:**

- a) Reliable
- b) Unreliable
- c) Fast
- d) Slow

**Answer: b**

*Explanation: As defined on Page 50, `udt_send()` stands for "unreliable data transfer".*

---

**112. The `deliver_data()` function in the rdt protocols is used to:**

- a) Send data to the network layer.
- b) Pass data up to the application layer.
- c) Check data for errors.
- d) Acknowledge received data.

**Answer: b**

*Explanation: Page 49 states: "When the rdt protocol wants to deliver data to the upper layer, it will do so by calling `deliver_data()`."*

---

**113. In rdt3.0, if an ACK is lost, what happens?**

- a) The receiver requests a retransmission.
- b) The sender's timer expires, and it retransmits the data packet.
- c) The connection is permanently broken.
- d) The network layer detects the loss and informs the transport layer.

**Answer: b**

*Explanation: This is the core mechanism of rdt3.0. The sender uses a timer to detect losses (whether of the data packet or the ACK) and retransmits.*

---

**114. The concept of pipelining improves performance by:**

- a) Reducing the transmission rate R.
- b) Allowing multiple packets to be in-flight before acknowledgments are received.
- c) Using a more efficient checksum algorithm.
- d) Compressing the application data.

**Answer: b**

*Explanation: Page 82 explains that pipelining allows the sender to send "multiple packets without waiting for acknowledgments," which fills the pipeline and increases utilization.*

---

**115. The receive window (rwnd) is advertised by the receiver to the sender in the:**

- a) Sequence Number field.
- b) Acknowledgment Number field.
- c) Receive Window field.
- d) Options field.

**Answer: c**

*Explanation: Page 112 identifies the "16-bit receive window field" in the TCP header, and Page 147 explains that the receiver places the rwnd value in this field.*

---

**116. The sequence of states for a server-side TCP during a normal connection teardown initiated by the client is:**

- a) ESTABLISHED -> FIN\_WAIT\_1 -> FIN\_WAIT\_2 -> TIME\_WAIT
- b) ESTABLISHED -> CLOSE\_WAIT -> LAST\_ACK -> CLOSED
- c) ESTABLISHED -> SYN\_RCVD -> LISTEN -> CLOSED
- d) ESTABLISHED -> SYN\_SENT -> CLOSED

**Answer: b**

*Explanation: Figure 3.42 on Page 161 shows the server's path during teardown: from ESTABLISHED, upon receiving a FIN, it goes to CLOSE\_WAIT, then after sending its own FIN, to LAST\_ACK, and finally to CLOSED upon receiving the final ACK.*

---

**117. The sequence of states for a client-side TCP during a normal connection teardown it initiates is:**

- a) ESTABLISHED -> FIN\_WAIT\_1 -> FIN\_WAIT\_2 -> TIME\_WAIT -> CLOSED
- b) ESTABLISHED -> CLOSE\_WAIT -> LAST\_ACK -> CLOSED
- c) ESTABLISHED -> SYN\_RCVD -> LISTEN -> CLOSED
- d) ESTABLISHED -> SYN\_SENT -> CLOSED

**Answer: a**

*Explanation: Figure 3.41 on Page 158 and the text on Pages 159-160 describe the client's path: ESTABLISHED -> FIN\_WAIT\_1 -> FIN\_WAIT\_2 -> TIME\_WAIT -> CLOSED.*

---

**118. A TCP sender is said to be "self-clocking" because:**

- a) It uses its own internal timer for all operations.
- b) The arrival of ACKs triggers the transmission of new data.
- c) It synchronizes its clock with the receiver.
- d) It operates independently of the application layer.

**Answer: b**

*Explanation: While not explicitly using the term "self-clocking," the concept is fundamental to how pipelined protocols like TCP and GBN work. The receipt of an ACK often allows the window to slide forward, enabling the sending of new data, thus making the ACKs "clock" the data flow.*

---

**119. The term "maximum transmission unit (MTU)" refers to the:**

- a) Maximum size of a transport-layer segment.
- b) Maximum size of a network-layer datagram.
- c) Maximum size of a link-layer frame.
- d) Maximum rate of a transmission link.

**Answer: c**

*Explanation: Page 108 defines MTU: "the largest link-layer frame that can be sent by the local sending host (the so-called maximum transmission unit, MTU)."*

---

**120. The MSS is typically set to:**

- a) The MTU.
- b) The MTU minus the TCP and IP header lengths.
- c) The receiver's advertised window.
- d) 1500 bytes.

**Answer: b**

*Explanation: Page 108 explains the calculation: "setting the MSS to ensure that a TCP segment... plus the TCP/IP header length... will fit into a single link-layer frame." This is  $MTU - (TCP\ Header + IP\ Header)$ .*

---

## Questions 121-130

**121. Which field in a TCP segment must be unique for each segment to avoid ambiguity in the presence of retransmissions?**

- a) Source Port
- b) Destination Port
- c) Sequence Number
- d) Checksum

**Answer: c**

*Explanation: The sequence number is the primary mechanism, introduced in rdt2.1 (Page 63), to allow the receiver to distinguish between new packets and retransmissions.*

---

**122. The event that causes the rdt1.0 receiver FSM to transition is:**

- a) rdt\_send(data) from above.
- b) A timeout.
- c) rdt\_rcv(packet) from below.
- d) Sending an ACK.

**Answer: c**

*Explanation: The FSM for the rdt1.0 receiver in Figure 3.9(b) on Page 52 has only one event: rdt\_rcv(packet) from below.*

---

**123. In rdt2.0, the sender retransmits a packet when it receives a:**

- a) ACK
- b) NAK
- c) New data from the upper layer
- d) Timeout

**Answer: b**

*Explanation: The sender FSM for rdt2.0 in Figure 3.10(a) on Page 58 shows that upon the event "rdt\_rcv(rcvpkt) && isNAK(rcvpkt)", the action is "udt\_send(sndpkt)" (i.e., retransmit).*

---

**124. In rdt3.0, the sender retransmits a packet when: (Select all that apply)**

- a) It receives a NAK.
- b) It receives a corrupted ACK.
- c) Its timer expires.
- d) It receives a duplicate ACK.

**Answer: a, b, c**

*Explanation: The rdt3.0 sender FSM in Figure 3.15 (Page 73) shows transitions for receiving a corrupt packet or an ACK for the wrong sequence number (which is like a NAK in rdt2.2), and for a timeout. Duplicate ACK handling is a TCP feature, not part of rdt3.0.*

---

**125. The primary cause of a timeout event in a reliable data transfer protocol is most likely:**

- a) A bit error in a packet.
- b) A lost packet.
- c) A reordered packet.
- d) A slow receiver.

**Answer: b**

*Explanation: While bit errors can cause timeouts if they corrupt the ACK, the primary cause discussed in the context of rdt3.0 and TCP is packet loss.*

---

**126. The purpose of the checksum in both UDP and TCP is to:**

- a) Correct bit errors.
- b) Detect bit errors.
- c) Compress the data.
- d) Encrypt the data.

**Answer: b**

*Explanation: As stated for UDP on Page 44, the checksum provides "error detection." The same is true for TCP.*

---

**127. A process can have multiple sockets to:**

- a) Increase the processing speed of the CPU.
- b) Communicate with multiple other processes simultaneously.
- c) Reduce the size of the transport-layer segment.
- d) Bypass the network layer.

**Answer: b**

*Explanation: The example on Page 17 (downloading Web pages, FTP, Telnet) implies that multiple sockets allow a single host to have multiple concurrent network connections.*

---

**128. The well-known port number for HTTP is:**

- a) 21
- b) 25
- c) 80
- d) 110

**Answer: c**

*Explanation: Page 20 lists HTTP as an example of a protocol using a well-known port number, and it's commonly known to be port 80.*

---

**129. The well-known port number for FTP is:**

- a) 21
- b) 25
- c) 80
- d) 110

**Answer: a**

*Explanation: Page 20 lists FTP as an example, and it's commonly known to be port 21.*

---

**130. In the context of TCP congestion control, doubling the timeout interval after a timeout is an example of:**

- a) Exponential backoff.
- b) Additive increase.
- c) Multiplicative decrease.
- d) Slow start.

**Answer: a**

*Explanation: Page 136 describes this behavior: "the intervals grow exponentially after each retransmission." This is the definition of exponential backoff.*

---

## Questions 131-140

**131. The TCP header option is used for: (Select all that apply)**

- a) Negotiating the Maximum Segment Size (MSS).
- b) Window scaling for high-speed networks.
- c) Time-stamping.
- d) Carrying application data.

**Answer: a, b, c**

*Explanation: Page 113 lists these uses for the options field.*

---

**132. The TCP segment structure allows for urgent data to be identified using the:**

- a) PSH flag and Receive Window field.
- b) URG flag and Urgent Data Pointer field.
- c) SYN flag and Sequence Number field.
- d) FIN flag and Acknowledgment Number field.

**Answer: b**

*Explanation: Page 114 states: "the URG bit is used... The location of the last byte of this urgent data is indicated by the 16-bit urgent data pointer field."*

---

**133. The initial value of the receiver's advertised window (rwnd) is typically:**

- a) 0
- b) The MSS
- c) The size of the receive buffer (RcvBuffer)
- d) The sequence number (client\_isn or server\_isn)

**Answer: c**

*Explanation: Page 147 states: "Initially, Host B sets  $rwnd = RcvBuffer$ ."*

---

**134. The sender must ensure that the amount of unacknowledged data is less than or equal to:**

- a) The congestion window (cwnd).
- b) The minimum of the congestion window (cwnd) and the receive window (rwnd).
- c) The receive window (rwnd).
- d) The send buffer size.

**Answer: b**

*Explanation: This is a key concept in TCP that combines flow control (rwnd) and congestion control (cwnd). The sender is limited by the minimum of these two windows. While the text on Page 149 only mentions rwnd in the flow control section, the concept of the congestion window is introduced later and is central to TCP's operation.*

---

**135. A fundamental difference between TCP and UDP is that TCP provides \_\_\_\_\_, while UDP does not.**

- a) Multiplexing/demultiplexing
- b) Error detection
- c) Connection-oriented, reliable byte-stream service
- d) A checksum for the payload

**Answer: c**

*Explanation: This is the core distinction summarized throughout the chapter, for example, on Page 11 and Page 14.*

---

**136. The "door" through which data passes from the network to a process, and vice versa, is called a:**

- a) Port
- b) Socket
- c) Buffer
- d) Segment

**Answer: b**

*Explanation: Page 17 defines a socket as "doors through which data passes from the network to the process and through which data passes from the process to the network."*

---

**137. In the Internet protocol stack, the transport layer resides directly above the:**

- a) Application Layer
- b) Network Layer
- c) Data Link Layer
- d) Physical Layer

**Answer: b**

*Explanation: Page 3 states the transport layer is "Residing between the application and network layers".*

---

**138. The protocol that provides a reliable, connection-oriented service is:**

- a) IP
- b) UDP
- c) TCP
- d) HTTP

**Answer: c**

*Explanation: Page 11 states: "TCP (Transmission Control Protocol), which provides a reliable, connection-oriented service".*

---

**139. The protocol that provides an unreliable, connectionless service is:**

- a) IP
- b) UDP
- c) TCP
- d) HTTP

**Answer: b**

*Explanation: Page 11 states: "UDP (User Datagram Protocol), which provides an unreliable, connectionless service".*

---

**140. The transport-layer packet in the Internet is called a:**

- a) Frame
- b) Datagram
- c) Segment

- d) Message

**Answer: c**

*Explanation: As explained on Page 11, the text uses the term "segment" for both TCP and UDP transport-layer packets.*

---

## Questions 141-150

**141. The network-layer packet in the Internet is called a:**

- a) Frame
- b) Datagram
- c) Segment
- d) Message

**Answer: b**

*Explanation: Page 11 states that the term "datagram" is reserved for the network-layer packet.*

---

**142. A host can have multiple simultaneous TCP connections to the same server process because each connection is identified by a unique:**

- a) Destination Port Number
- b) Source Port Number
- c) Four-tuple (Source IP, Source Port, Destination IP, Destination Port)
- d) Sequence Number

**Answer: c**

*Explanation: As explained on Page 27, the four-tuple uniquely identifies a TCP connection. Multiple clients can use the same server port (e.g., 80) but have different source IPs and/or source ports.*

---

**143. The concept of a "sliding window" is used in protocols to:**

- a) Manage flow control and allow multiple in-flight packets.
- b) Detect errors in the data stream.
- c) Establish connections reliably.
- d) Tear down connections gracefully.

**Answer: a**

*Explanation: The sliding window protocol (like GBN) is introduced to manage the pipeline of unacknowledged packets, which is directly related to flow control and utilization.*

---

**144. The rdt\_send() function is called by:**

- a) The lower layer (network layer) to deliver a packet to the transport layer.
- b) The upper layer (application layer) to pass data to the transport layer.
- c) The timer interrupt handler.
- d) The receiving side of the transport layer.

**Answer: b**

*Explanation: Page 49 states: "The sending side of the data transfer protocol will be invoked from above by a call to rdt\_send()."*

---

**145. The rdt\_rcv() function is called when:**

- a) The application has data to send.
- b) A packet arrives from the network layer.
- c) The timer expires.
- d) An acknowledgment needs to be sent.

**Answer: b**

*Explanation: Page 49 states: "On the receiving side, rdt\_rcv() will be called when a packet arrives from the receiving side of the channel."*

---

**146. In the GBN protocol, if an ACK for a packet is lost, what happens?**

- a) Only that packet is eventually retransmitted when its timer expires.
- b) The sender's window slides forward, and no retransmission occurs.
- c) The sender retransmits that packet and all subsequent packets when a timeout occurs.
- d) The receiver sends a NAK for the lost ACK.

**Answer: c**

*Explanation: This is the core "Go-Back-N" behavior. Since ACKs are cumulative, the loss of an ACK might not be critical if a later ACK is received. But if a timeout occurs for the oldest unacknowledged packet, the sender retransmits that packet and all newer ones in the window.*

---

**147. In the SR protocol, the sender and receiver windows:**

- a) Always have the same size.
- b) Are always of size 1.
- c) Do not necessarily coincide.
- d) Are not used.

**Answer: c**

*Explanation: In SR, the sender and receiver maintain their own windows. The receiver's window reflects what it can accept, while the sender's window tracks what has been sent and acknowledged. They are related but not always identical, especially with losses.*

---

**148. The TCP sender's effective window is determined by:**

- a) The receiver's advertised window (rwnd) only.
- b) The congestion window (cwnd) only.
- c) The minimum of the receiver's advertised window (rwnd) and the congestion window (cwnd).
- d) The maximum of the receiver's advertised window (rwnd) and the congestion window (cwnd).

**Answer: c**

*Explanation: As mentioned in the explanation for question 134, the sender is limited by both flow control (rwnd) and congestion control (cwnd), taking the minimum of the two.*

---

**149. A key difference between TCP and GBN is that TCP:**

- a) Uses cumulative acknowledgments exclusively.
- b) Always retransmits all unacknowledged packets on a timeout.
- c) Often buffers out-of-order segments at the receiver.
- d) Has a fixed window size.

**Answer: c**

*Explanation: Page 141-142 highlight this difference: "Many TCP implementations will buffer correctly received but out-of-order segments," whereas a GBN receiver discards them.*

---

**150. The primary motivation for using UDP for multimedia applications is:**

- a) Reliability
- b) Congestion control

- c) Low delay and tolerance for minor loss
- d) Large header size

**Answer: c**

*Explanation: Page 39 explains that multimedia applications "can tolerate a small amount of packet loss" and "react very poorly to TCP's congestion control," making UDP attractive for its lower delay.*

---

## Questions 151-160

**151. The problem of "premature timeout" in rdt3.0 can lead to:**

- a) Permanent connection loss.
- b) Duplicate packets.
- c) Corrupted data.
- d) Faster transmission.

**Answer: b**

*Explanation: Page 72 discusses this: if a packet is delayed, the sender may retransmit, "introducing the possibility of duplicate data packets."*

---

**152. The sequence number in rdt3.0 allows the receiver to:**

- a) Detect bit errors.
- b) Detect lost packets.
- c) Detect duplicate packets.
- d) Control the flow of packets.

**Answer: c**

*Explanation: The sequence number's primary role, as introduced in rdt2.1, is to allow the receiver to distinguish new packets from retransmissions (duplicates).*

---

**153. The length field in the UDP header specifies the length of the:**

- a) Entire UDP segment (header plus data).
- b) UDP header only.
- c) UDP data field only.
- d) Underlying IP datagram.

**Answer: a**

*Explanation: Page 43 states: "The length field specifies the number of bytes in the UDP segment (header plus data)."*

---

**154. The Internet checksum is calculated over the:**

- a) UDP header and data only.
- b) TCP header and data only.
- c) UDP/TCP header, data, and some fields from the IP header (pseudo-header).
- d) Entire IP datagram.

**Answer: c**

*Explanation: Page 43 mentions for UDP: "the checksum is also calculated over a few of the fields in the IP header in addition to the UDP segment." This concept of a "pseudo-header" also applies to TCP.*

---

**155. The TCP state "LISTEN" is typically associated with the:**

- a) Client waiting for a connection request.
- b) Server waiting for a connection request.
- c) Client after sending a SYN.
- d) Server after sending a SYNACK.

**Answer: b**

*Explanation: Figure 3.42 on Page 161 shows the server in the LISTEN state after the "Server application creates a listen socket".*

---

**156. The main purpose of the transport layer is to provide \_\_\_\_\_ communication, while the network layer provides \_\_\_\_\_ communication.**

- a) host-to-host; process-to-process
- b) process-to-process; host-to-host
- c) end-to-end; hop-by-hop
- d) reliable; best-effort

**Answer: b**

*Explanation: This is the fundamental distinction reiterated on Page 8.*

---

**157. In a high-performing Web server, multiple client connections may be handled by:**

- a) Multiple processes, each with its own socket.
- b) A single process with multiple threads, each with its own connection socket.
- c) A single process with a single socket for all connections.
- d) Either a or b.

**Answer: d**

*Explanation: Page 30 describes both models: "a Web server that spawns a new process for each connection" and "today's high-performing Web servers often use only one process, and create a new thread with a new connection socket for each new client connection."*

---

**158. For non-persistent HTTP, a new TCP connection (and socket) is created for:**

- a) The entire duration of the user's visit to the website.
- b) Each request/response pair.
- c) Only the initial request.
- d) Only the responses containing images.

**Answer: b**

*Explanation: Page 30 states: "if the client and server use non-persistent HTTP, then a new TCP connection is created and closed for every request/response".*

---

**159. The problem of "wasting" bandwidth by retransmitting correctly received packets is most severe in which protocol when the window size is large?**

- a) Stop-and-Wait (rdt3.0)
- b) Go-Back-N (GBN)
- c) Selective Repeat (SR)
- d) TCP with fast retransmit

**Answer: b**

*Explanation: Page 98 identifies this as the key problem that SR solves: "A single packet error can thus cause GBN to retransmit a large number of packets, many unnecessarily."*

---

**160. The concept of "pipelining" requires that the range of sequence numbers must be:**

- a) Decreased
- b) Increased
- c) The same as for stop-and-wait

- d) Exactly 2

**Answer: b**

*Explanation: Page 84 lists as the first consequence of pipelining: "The range of sequence numbers must be increased".*

---

## Questions 161-170

**161. The event that causes a transition from the "Wait for ACK 0" state to the "Wait for call 1 from above" state in the rdt3.0 sender FSM is:**

- a) rdt\_send(data)
- b) Timeout
- c) rdt\_rcv(rcvpkt) && corrupt(rcvpkt)
- d) rdt\_rcv(rcvpkt) && notcorrupt(rcvpkt) && isACK(rcvpkt,1)

**Answer: d**

*Explanation: Looking at the rdt3.0 sender FSM in Figure 3.15 (Page 73), the transition from the "Wait for ACK 0" state (left side) back to the "Wait for call 1 from above" state (right side) occurs when a valid ACK for sequence number 1 is received. This acknowledges the packet with seq=0, so the sender can move on to sending packets with seq=1.*

---

**162. The "Finite State Machine (FSM)" is used in this chapter to model the behavior of:**

- a) The network layer.
- b) The transport layer protocols.
- c) The application layer processes.
- d) The physical layer links.

**Answer: b**

*Explanation: The FSM descriptions are used extensively in Section 3.4 to describe the rdt protocols, which are transport-layer mechanisms.*

---

**163. The term "segment" is used in the Internet for the transport-layer packet because:**

- a) It is a segment of the application data.
- b) It is a segment of the network path.
- c) It is a segment of the link-layer frame.
- d) It is a segment of the entire message.

**Answer: a**

*Explanation: The transport layer takes the application message and breaks it into chunks (segments) for transmission, as described on Page 6.*

---

**164. The main role of the transport layer is to provide logical communication between \_\_\_\_\_, while the main role of the network layer is to provide logical communication between \_\_\_\_\_.**

- a) applications; hosts
- b) processes; hosts
- c) hosts; processes
- d) sockets; ports

**Answer: b**

*Explanation: This is the core concept from Page 8.*

---

**165. A TCP segment is encapsulated in:**

- a) An IP datagram.
- b) A UDP segment.
- c) An Ethernet frame.
- d) An HTTP message.

**Answer: a**

*Explanation: Page 109 states: "The segments are passed down to the network layer, where they are separately encapsulated within network-layer IP datagrams."*

---

**166. The UDP header is \_\_\_\_\_ bytes long.**

- a) 8
- b) 20
- c) 40
- d) Variable

**Answer: a**

*Explanation: Page 43 states the UDP header has four 2-byte fields, so  $4 * 2 = 8$  bytes.*

---

**167. The typical TCP header is \_\_\_\_\_ bytes long.**

- a) 8
- b) 20
- c) 40
- d) Variable

**Answer: b**

*Explanation: Page 110 states: "TCP header is typically 20 bytes".*

---

**168. The TCP header can be longer than 20 bytes due to the:**

- a) Data field.
- b) Options field.
- c) Checksum field.
- d) Window field.

**Answer: b**

*Explanation: Page 112 states: "The TCP header can be of variable length due to the TCP options field."*

---

**169. The TCP flag bits used for connection establishment and teardown are:**

- a) ACK and PSH
- b) SYN and FIN
- c) RST and URG
- d) CWR and ECE

**Answer: b**

*Explanation: Page 113 states: "The RST, SYN, and FIN bits are used for connection setup and teardown".*

---

**170. The fundamental problem that reliable data transfer protocols solve is:**

- a) How to route packets through the network.
- b) How to convert digital signals to analog.
- c) How two entities can communicate reliably over an unreliable medium.
- d) How to compress data for efficient transmission.

**Answer: c**

*Explanation: Page 3 identifies this as "one of the most fundamental problems in computer networking".*

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**Wish you all the best!**  
**Fathy.**